

Assessment Rubric: PostgreSQL Mastery Capstone

Detailed Evaluation Criteria

This rubric is used for the Week 12 final assessment, the capstone project review, and weekly self-evaluation. Each criterion has three performance levels: Developing (1), Proficient (2), and Expert (3).

Dimension 1: Schema Design (25%)

1.1 Normalization (33% of dimension)

Level	Score	Description
Expert	3	Schema is in 3NF. Any denormalization is documented with clear business justification and query evidence. No update anomalies possible.
Proficient	2	Schema is in 2NF. At most 1-2 transitive dependencies present. No obvious redundancy.
Developing	1	Multiple normalization violations. Repeated data in multiple tables. Potential for insert/update/delete anomalies.

1.2 Constraints and Data Integrity (33% of dimension)

Level	Score	Description
Expert	3	All business rules enforced at DB level. CHECK constraints on all domain-restricted columns. NOT NULL on all required fields. UNIQUE where appropriate. Custom DOMAINS used for reusable validation. Computed columns for derived values.
Proficient	2	Primary and foreign keys all correct. Most NOT NULL constraints present. CHECK constraints on critical columns.
Developing	1	Primary keys present but foreign keys missing or incorrect. Few CHECK constraints. Business rules only enforced in application code.

1.3 Naming and Conventions (17% of dimension)

Level	Score	Description
Expert	3	Consistent snake_case throughout. Tables are plural. Boolean columns use is_/has_ prefix. Timestamps use _at suffix. Foreign keys follow table_id pattern. Acronyms expanded.
Proficient	2	Mostly consistent. Minor inconsistencies in a few table or column names.
Developing	1	Mixed conventions. CamelCase and snake_case in same schema. Unclear abbreviations.

1.4 Scalability Consideration (17% of dimension)

Level	Score	Description
Expert	3	Partitioning applied to high-volume tables. Soft deletes on appropriate tables. Audit trail tables present. Denormalized counters documented. Schema would survive 100x data growth.
Proficient	2	Basic scalability patterns applied. Partitioning considered. No obvious growth-blockers.
Developing	1	Schema would become unmanageable at 10x growth. Missing partitioning on time-series tables.

Dimension 2: SQL Query Quality (25%)

2.1 Correctness (40% of dimension)

Level	Score	Description
Expert	3	All queries return exactly correct results. NULL handling is explicit. Edge cases (empty set, all-NULL input) handled. Queries verified with test data covering boundary conditions.
Proficient	2	Most queries correct. May have minor NULL-handling gaps. Correct for happy-path scenarios.
Developing	1	Several queries return incorrect results. NULL semantics misunderstood. JOINS produce cartesian products in some cases.

2.2 Advanced SQL Usage (30% of dimension)

Level	Score	Description
Expert	3	Window functions used where appropriate (not just WHERE + subquery). Recursive CTEs for hierarchies. FILTER clause for conditional aggregation. LATERAL joins where applicable. No unnecessary subqueries where CTEs are cleaner.
Proficient	2	Window functions used in at least 3 queries. CTEs used for readability. Basic aggregations correct.
Developing	1	Only basic SELECT, JOIN, GROUP BY. No window functions. All aggregations with simple subqueries even when cleaner options exist.

2.3 Readability and Documentation (30% of dimension)

Level	Score	Description
Expert	3	Every query has a comment explaining business context. Complex expressions explained inline. CTE names are self-documenting. Consistent formatting. Alias columns with meaningful names.
Proficient	2	Most queries commented. Formatting is consistent. CTE names meaningful.

Developing	1	Queries lack comments. SELECT * used. Magic numbers with no explanation.
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Dimension 3: PL/pgSQL Functions and Triggers (20%)

3.1 Function Correctness and Business Logic (40% of dimension)

Level	Score	Description
Expert	3	Functions implement all specified business rules correctly. Return types are appropriate. Functions tested with valid, invalid, and boundary inputs. Atomic operations use transactions correctly.
Proficient	2	Functions implement core logic correctly. Most edge cases handled.
Developing	1	Functions work for happy path only. Errors in edge cases. Logic bugs present.

3.2 Error Handling and Robustness (30% of dimension)

Level	Score	Description
Expert	3	All functions raise meaningful EXCEPTION messages with context (e.g., RAISE EXCEPTION 'Insufficient funds for account %: balance=%, requested=%', account_id, balance, amount). NULL inputs validated. Foreign key violations handled. Deadlock-safe using consistent lock ordering.
Proficient	2	Most error conditions raise exceptions. Basic NULL checks.
Developing	1	Errors surface as cryptic system errors. No explicit error handling.

3.3 Trigger Design (30% of dimension)

Level	Score	Description
Expert	3	Triggers used for cross-row enforcement, audit logging, and computed column maintenance. Triggers are correctly scoped (BEFORE vs AFTER, ROW vs STATEMENT). Triggers do not duplicate logic already handled by constraints.
Proficient	2	At least 2 triggers with correct timing and scope. Trigger logic correct.
Developing	1	Triggers present but wrong timing or scope. Logic errors.

Dimension 4: Performance Optimization (20%)

4.1 Index Strategy (50% of dimension)

Level	Score	Description
Expert	3	Correct index type for each query pattern (B-Tree, GIN, BRIN, partial). Covering indexes for index-only scans. Partial indexes for filtered queries. No redundant or unused indexes. Index sizes documented.

Proficient	2	B-Tree indexes on all foreign keys and common filter columns. At least 1 partial index.
Developing	1	Few or no indexes beyond primary key. Foreign keys not indexed.

4.2 EXPLAIN ANALYZE Usage (30% of dimension)

Level	Score	Description
Expert	3	EXPLAIN ANALYZE run on all major queries. Before/after comparison documented for optimizations. Can explain every node in execution plan. Buffer usage analyzed with BUFFERS option.
Proficient	2	EXPLAIN ANALYZE used for at least 3 queries. Index usage verified. Seq scans investigated.
Developing	1	EXPLAIN never used or used without understanding output.

4.3 Materialized Views and Caching (20% of dimension)

Level	Score	Description
Expert	3	Materialized views for expensive aggregation queries. Refresh strategy documented. CONCURRENTLY used where appropriate. Partial refresh considered.
Proficient	2	At least 1 materialized view with index and refresh plan.
Developing	1	No materialized views. Expensive queries run on every request.

Dimension 5: Documentation and Communication (10%)

5.1 Code Comments and README (50% of dimension)

Level	Score	Description
Expert	3	Every table has a COMMENT ON TABLE. Complex columns commented. README explains architecture, design decisions, and how to run the project. Architecture Decision Records for non-obvious choices.
Proficient	2	Most tables commented. README present with setup instructions.
Developing	1	No comments. No README. Code is not self-documenting.

5.2 Verbal Explanation (50% of dimension)

Level	Score	Description
Expert	3	Can explain any design decision without notes. Proactively discusses trade-offs. Acknowledges what they would do differently. Anticipates follow-up questions.
Proficient	2	Can explain most decisions clearly. Some notes needed. Handles follow-ups well.

Developing	1	Struggles to explain decisions. Cannot articulate trade-offs.
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Scoring Summary

Calculate your score for each dimension:

Dimension 1 (Schema Design, 25%)

1.1 Normalization: $\frac{\quad}{3} \times 0.33 \times 25 = \quad$
 1.2 Constraints: $\frac{\quad}{3} \times 0.33 \times 25 = \quad$
 1.3 Naming: $\frac{\quad}{3} \times 0.17 \times 25 = \quad$
 1.4 Scalability: $\frac{\quad}{3} \times 0.17 \times 25 = \quad$
 Subtotal: $\quad / 25$

Dimension 2 (Queries, 25%)

2.1 Correctness: $\frac{\quad}{3} \times 0.40 \times 25 = \quad$
 2.2 Advanced SQL: $\frac{\quad}{3} \times 0.30 \times 25 = \quad$
 2.3 Readability: $\frac{\quad}{3} \times 0.30 \times 25 = \quad$
 Subtotal: $\quad / 25$

Dimension 3 (Functions, 20%)

3.1 Business Logic: $\frac{\quad}{3} \times 0.40 \times 20 = \quad$
 3.2 Error Handling: $\frac{\quad}{3} \times 0.30 \times 20 = \quad$
 3.3 Triggers: $\frac{\quad}{3} \times 0.30 \times 20 = \quad$
 Subtotal: $\quad / 20$

Dimension 4 (Performance, 20%)

4.1 Indexes: $\frac{\quad}{3} \times 0.50 \times 20 = \quad$
 4.2 EXPLAIN: $\frac{\quad}{3} \times 0.30 \times 20 = \quad$
 4.3 Materialized Views: $\frac{\quad}{3} \times 0.20 \times 20 = \quad$
 Subtotal: $\quad / 20$

Dimension 5 (Documentation, 10%)

5.1 Comments/README: $\frac{\quad}{3} \times 0.50 \times 10 = \quad$
 5.2 Verbal: $\frac{\quad}{3} \times 0.50 \times 10 = \quad$
 Subtotal: $\quad / 10$

TOTAL: $\quad / 100$

Performance Bands

Score	Band	Career Readiness
90-100	Expert	Senior DB engineer / DBA roles
75-89	Proficient	Mid-level backend/data engineering
60-74	Competent	Junior roles with continued growth
45-59	Developing	Review Weeks 5-8 before applying
Below 45	Foundational	Return to Week 1 and repeat the program

Weekly Self-Assessment Template

Use this at the end of each week to track your progress:

Week: ____ Date: ____

What I learned this week (3 things):
1.
2.
3.

What I'm still confused about (1-2 things):
1.
2.

Evidence of learning (SQL I wrote, project I built):

Score on this week's mock interview questions: ____/10

Goal for next week:

Revision Tracker

Mark each dimension when you reach "Proficient" and again when you reach "Expert":

Dimension	Proficient Date	Expert Date
Schema Design		
SQL Query Quality		
PL/pgSQL Functions		
Performance		
Documentation		

When all five dimensions show an Expert date, you have completed the program.